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Functional recovery following brain and spinal cord injuries and neurodegenerative diseases is likely to require the transplantation of exogenous neural cells and tissues, since the mammalian central nervous system (CNS) has little capacity for self-repair. However, neural cell or tissue transplantation is limited by the lack of tissue donors and the low survival rate of grafted cells. There is a critical need for an alternative approach to building biological substitutes, such as three-dimensional (3D) in vitro culture of neural cells to repair or replace the function of damaged nerve tissues. Our strategy is to place dissociated neural stem cells onto naturally-derived or synthetic hydrogel scaffolds and to culture the resulting cell-hydrogel constructs in a RWV bioreactor. The in vitro system we propose here includes three components: 1) primary neural stem cells isolated from embryonic rat cortical neuroepithelium; 2) naturally-derived and synthetic or biosynthetic hydrogels modified by brain ECM/adhesive proteins and loaded with growth factors, that promote cell survival, growth and differentiation; and 3) RWV bioreactors that offer an optimized microgravity environment and regulate tissue development. Our working hypothesis is that rotating bioreactors offer relatively well-controlled process

Task Description:

parameters and reduced hydrodynamic forces that may regulate neural stem cell proliferation, neurogenesis and synaptic formation. Rotating bioreactors create low shear stress and randomized gravitational vectors that optimize cell recruitment. They also provide high supply rates of nutrients and gases that are expected to improve neural tissue reconstruction in our in vitro system. The proposed cell-hydrogel-bioreactor system may allow us to study and thereby control neural tissue growth and neuronal network formation under simulated microgravity conditions. A successfully engineered nerve tissue may have potential use in tissue replacement therapy for brain and spinal cord injuries and neurodegenerative diseases. Our studies of neurogenesis and neuronal network formation in the cell-hydrogel-bioreactor system may lead to a tissue model for potential space experiments in the future. The long-term objectives of this project are to understand how microgravity and reduced hydrodynamic stress affect neural tissue generation from neural stem cells, and to develop a general approach to engineering functional neuronal networks for potential use in neural tissue replacement therapy. To accomplish these goals, four specific aims are proposed. Aim 1 is to develop or optimize methods to use naturally derived collagen and synthetic hydrogels such as biodegradable polymer, Poly(lactic acid) modified by extracellular matrix proteins for scaffold preparation. In Aim 2, we seek to immobilize neural stem cells and progenitors in the hydrogel scaffolds. We will characterize the resulting cell-hydrogel interface by examining cell adhesion and integrin-mediated signal transduction. Aim 3 is to characterize neurogenesis in simulated microgravity. We will examine whether hydrodynamic conditions in RWV bioreactors affect neural stem cell proliferation. We will then assess the effects of reduced hydrodynamic stress and microgravity on neural cell lineage progression and neuronal production. In Aim 4, we will characterize synaptogenesis and functional neuronal network formation in simulated microgravity. We reported the first engineered construct resembling embryonic brain tissue generated by the stem cell-collagen-bioreactor culture system. This novel strategy constructs multi-layered neural tissue whose stratification resembles that of the embryonic rat cerebral cortex. There are at least three major advantages to this strategy for neural tissue engineering. First, the use of neural stem and progenitor cells, which are capable of both self-renewal and phenotypic differentiation into all types of neural cells, has the potential for supplying different cell types under controlled conditions. Second, the use of collagen, which is the most widespread ECM protein in mammalian tissues, provides a scaffold for cell growth in three-dimensions. The matrix is useful for the growth of anchorage-dependent cells like neural stem and progenitor cells into neuronal circuits and networks. Third, RWV bioreactors provide simulated microgravity that creates low fluid shear stress and randomized gravitational vectors. Compared to cultures under static 1g condition, the RWV-generated tissues contain cells with significantly higher rates of proliferation and survival. The majority of cells aggregating into spheres in static cultures died in 4-5 weeks while RWV cultures promoted compact aggregates, which became transformed into a multi-layered tissue without a necrotic core. Thus far, the functional engineered tissue-like constructs can be maintained in vitro for up to 9 weeks. This method demonstrates the promise of utility of tissue engineered constructs not only for in vitro studies of neural stem cell biology, but also for a tissue replacement strategy with new functional brain-like tissues.
The novel approach described here would be amenable to build different neural tissue models depending on controlled conditions. For example, a combination of bFGF, EGF and LIF would be a potent synergistic stimulation of neural stem cell proliferation, that generates a continuously proliferating population of immature cells. In order to generate more excitable neurons and active synaptic activity, an early withdrawal of bFGF and addition of BDNF are necessary. Furthermore, the method described herein would be amenable to build other types of tissue models. As shown previously, a cell-polymer-bioreactor system was used to promote chondrogenesis and cardiac tissue formation. This strategy could also be used to generate functional CNS tissue constructs derived from neural stem and progenitor cells from a variety of species including humans. Thus, in the future the in vitro model could be scaled-up for clinical use. In addition, this method can be applied to the generation of a cheap and reproducible in vitro system with greater control of variables for the testing of drugs and for detection of environmental toxins.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Work in the FY04 focused on developing a neural stem cell-collagen-bioreactor culture system for engineering living neural tissue in vitro. This system provides naturally derived polymer scaffolds, simulated microgravity and soluble factors that supply the physical and chemical cues to guide proliferation and differentiation of neural stem and progenitor cells isolated from the embryonic mammalian brain. This report describes the first engineered construct resembling embryonic brain tissue generated by using the system. In the engineered construct, neural stem and progenitor cells actively proliferate and differentiate into neurons, astrocytes and oligodendrocytes, and form a complex three-layered structure that superficially emulates the cerebral cortex of the embryonic brain.
Engineering Neural Tissue

The novel strategy for engineering living neural tissue in vitro involves entrapment of neural stem and progenitor cells into a collagen gel, which is then maintained for weeks in a RWV bioreactor. The in vitro system we developed includes three components: 1) primary neural stem and progenitor cells isolated from the embryonic rat cortical neuroepithelium at the beginning of neurogenesis; 2) type I collagen as the ECM protein to promote cell attachment, survival, growth and differentiation in three dimensions; and 3) NASA-designed RWV bioreactors that rotate around its central spin axis at a speed of 10-15 rpm so as to maintain the forming tissue in a state of continuous free-fall, which offers an optimized microgravity environment to regulate tissue development. The primary neural stem and progenitor cells were cultured in the cell-collagen-bioreactor system for up to 9 weeks in a serum-free medium containing basic fibroblast growth factor (bFGF) and together with brain derived neurotrophic factor (BDNF).

Cell Organization Emulates Native Cortical Tissue The collagen-entrapped neural stem and progenitor cells actively proliferated and differentiated into the major neural phenotypes composing the central nervous system, including neurons, astrocytes and oligodendrocytes. The cell-collagen constructs cultured in RWV bioreactors gradually compacted 4-5 fold and became elliptically-shaped, tissue-like structures. The compact size of the tissue contrasts with the progressive expansion of neuroepithelial cells, which have been cultured under static conditions (without rotation). Cells grown in the latter environment expand faster but many of the cells do not survive after 4 weeks. Inspection of the tissue-like structure consistently revealed multiple layers of cells, which were populated by different phenotypes. Immunostaining of the intact construct or transverse sections through the construct revealed the different phenotypes and their distributions. The external layer composing the surface of the tissue (layer I, interfacing with the culture medium) consisted of a thin sheet of progenitor cells (nestin+, vimentin+ and PCNA+). The deeper layer (layer II) was composed of a thicker uneven and continuous mass of differentiating neurons (TuJ1+), astrocytes (GFAP+) and oligodendrocytes (O4+). In the center core (layer III) there were small scattered aggregates of neurons (TuJ1+) and glial

Task Progress:	(GFAP+ or O4+) cells distributed throughout collagen matrix. The multiple layers of different phenotypes composing the RWV-generated tissue are reminiscent of the developing rat cerebral cortex. Proliferating neural stem and progenitor cells form the ventricular zone/subventricular zone (VZ/SVZ) while differentiating neuronal and glial phenotypes are found in the cortical plate/subplate (CP/SP) regions. Thus, the outside of the tissue (layer I) corresponds to the VZ/SVZ, while the layer II emulates the CP/SP regions. Layer III may be similar to the intermediate zone (IZ), which is relatively cell poor. However, in the primary tissue the IZ lies between the VZ/SVZ and the CP/SP. The growth conditions imposed here may have displaced this aspect of normal architecture. Alternatively, layer III is composed of differentiating phenotypes that have migrated away from layer II. In this case, the developing tissue is compacted by the rotational forces so that although the tissue is polarized into a neuroepithelial sheet of neural progenitor cells (layer I) and a multi-layer region of differentiating phenotypes (layer II), it is missing the intermediate zone. Finally, progenitor cells in layer I and some cells in layer II showed an ability to synthesize fibronectin, a unique ECM protein in the CNS. Distributions of Functional Transmitter Receptors and Na+ Channels, and Synaptic Vesicle Recycling in Engineered Tissue Resemble That Developing in vivo We investigated the distributions of functional receptors for neurotransmitters and Na+ channels among cells composing layers I and II using Ca2+ imaging. The layers were carefully separated before the cells were loaded with Ca2+ indicator dye. Sequential exposure to agonists at four major neurotransmitter receptors (cholinergic, purinergic, GABAergic and glutamatergic) revealed that the vast majority of cells (86%) in layer I responded to ATP, which activates purinergic receptors. The rise in cytosolic Ca2+ (Ca2+c) was high in amplitude and transient, relaxing to the baseline during the exposure. About half (55%) the cells responded to carbarchol, an agonist at muscarinic cholinergic receptors with a lower amplitude Ca2+c response, which was also transient. Few cells responded to the other agonists and all of these latter cells were clustered together. In contrast, few cells in layer II responded to ATP and those that did not exhibited low amplitude responses. The majority of cells responded to each of the other agonists. These differential results on cells in layers I and II parallel previously published results on proliferating precursor cells in the VZ/SVZ and differentiating neurons in the CP/SP. Thus, the two cell layers composing the engineered tissue exhibit properties similar to those developing in vivo. The second functional assay using an endocytotic marker (FM1-43) showed that cells in layer II, but not those in layer I, were stained with FM1-43, which has been used to label recycling synaptic vesicles in neuronal networks. FM1-43 fluorescence intensity was reduced upon K+ stimulation, demonstrating synaptic vesicle recycling at active neuronal circuits.

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